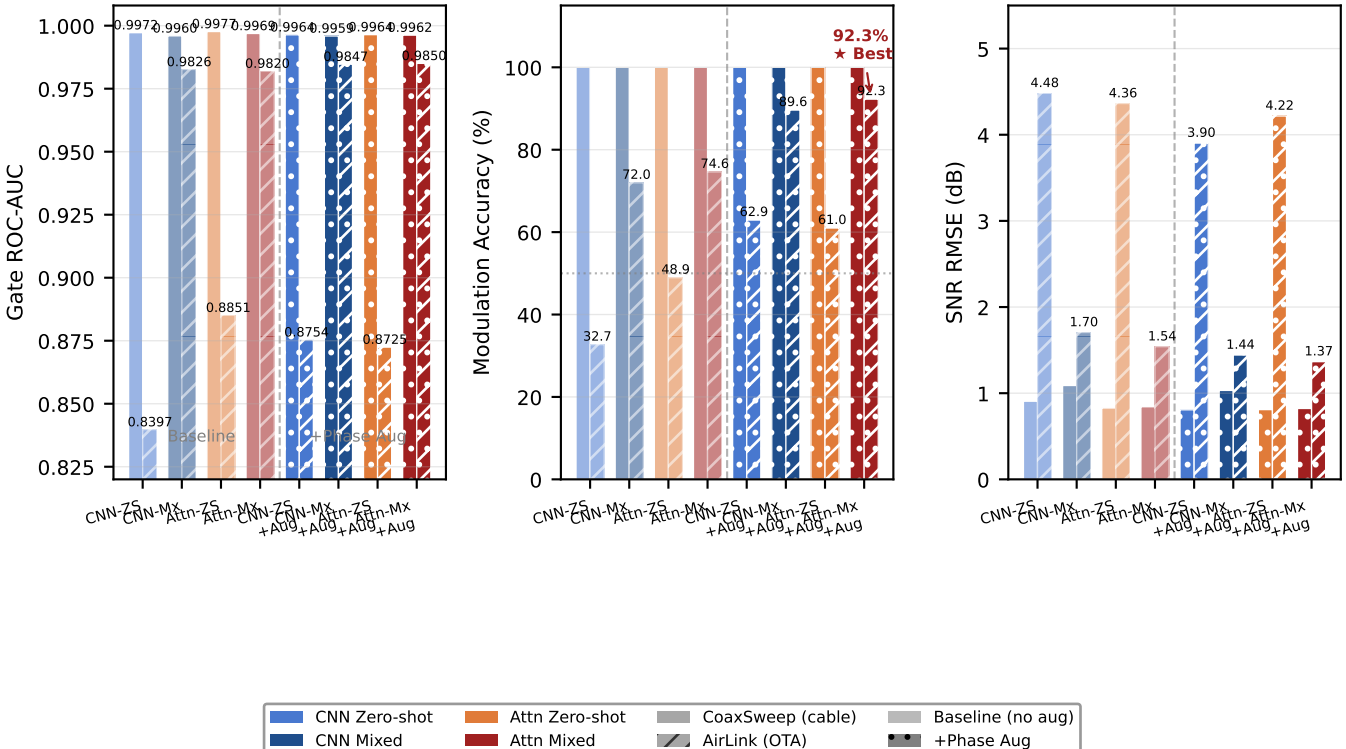

(a) Gate AUC
Gate ROC-AUC
1.000
0.975
0.950
0.925
0.900
0.875
0.850
0.825
0.9972
0.8397
0.9960
0.9826
0.9977
0.8851
0.9969
0.9820
0.9964
0.8754
0.9959
0.9847
0.9964
0.8725
0.9962
0.9850
Baseline
+Phase Aug
CNN-ZS
CNN-Mx
Attn-ZS
Attn-Mx
CNN-ZS +Aug
CNN-Mx +Aug
Attn-ZS +Aug
Attn-Mx +Aug
(b) Mod Accuracy
Modulation Accuracy (%)
100
80
60
40
20
0
92.3%
★ Best
32.7
72.0
48.9
74.6
62.9
89.6
61.0
92.3
(c) SNR RMSE
SNR RMSE (dB)
5
4
3
2
1
0
4.48
1.70
4.36
1.54
3.90
1.44
4.22
1.37
CNN Zero-shot
CNN Mixed
Attn Zero-shot
Attn Mixed
CoaxSweep (cable)
AirLink (OTA)
Baseline (no aug)
+Phase Aug